

Project Design Phase Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-2391
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

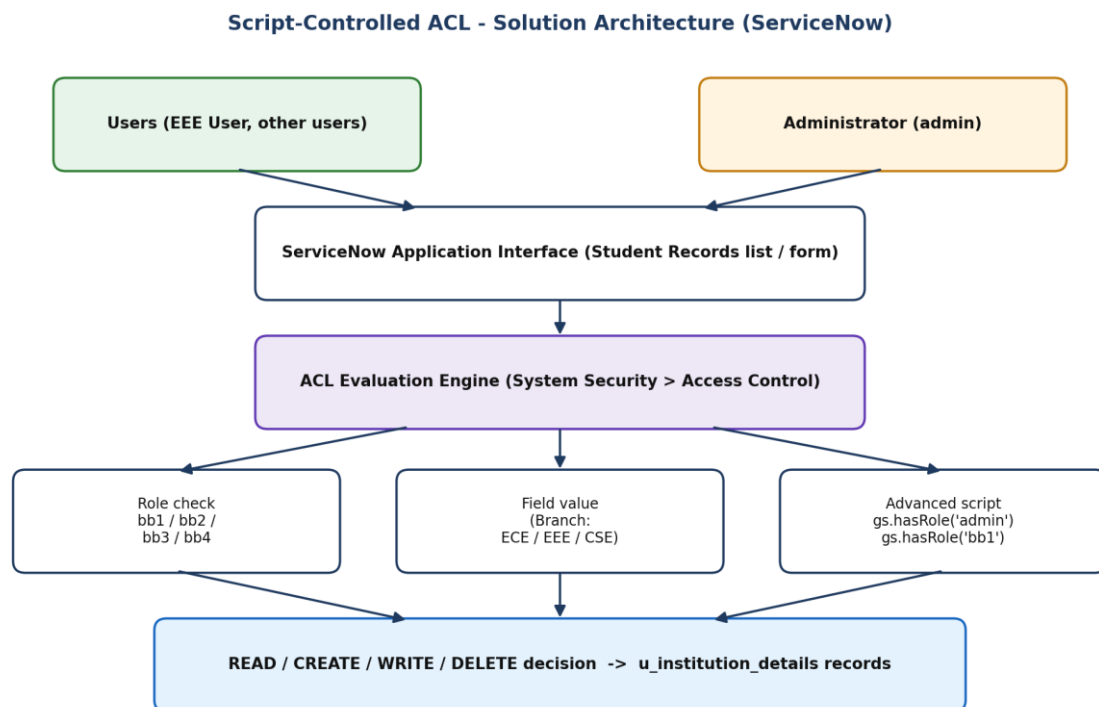


Figure 1: Architecture and access-decision flow of the Script-Controlled ACL solution on ServiceNow

Architecture Description:

- **Business problem:** restrict access to student records in a shared table based on user role and field value, with separate control of READ, CREATE, WRITE and DELETE.
- **User and role layer:** the EEE User holds custom roles bb1, bb2, bb3 and bb4; the administrator holds the admin role and has full access.
- **Data layer:** custom table u_institution_details (Institution Details) with Student Roll Number, Student Name, Faculty Name, Branch (ECE/EEE/CSE), Email, Phone Number and Description.

- **Access-control layer:** READ ACL (advanced, with script: admin and bb1 return true, all others false); CREATE ACL requires bb2; WRITE ACL requires bb3; DELETE ACL requires bb4.
- **Flow:** User or Administrator → ServiceNow interface → ACL evaluation → role / field-value / script check → allow or deny → record operation on u_institution_details.
- **Verification:** access is tested by impersonating users with different role combinations (bb1; bb1+bb2; bb1+bb2+bb3; bb1+bb2+bb3+bb4; administrator).

Development Phases:

- Phase 1 – User, role and custom table configuration.
- Phase 2 – Script-controlled READ ACL.
- Phase 3 – CREATE, WRITE and DELETE ACLs.
- Phase 4 – Testing through impersonation and documentation.

Reference:

<https://github.com/snekharamiah/Script-Controlled-ACL>